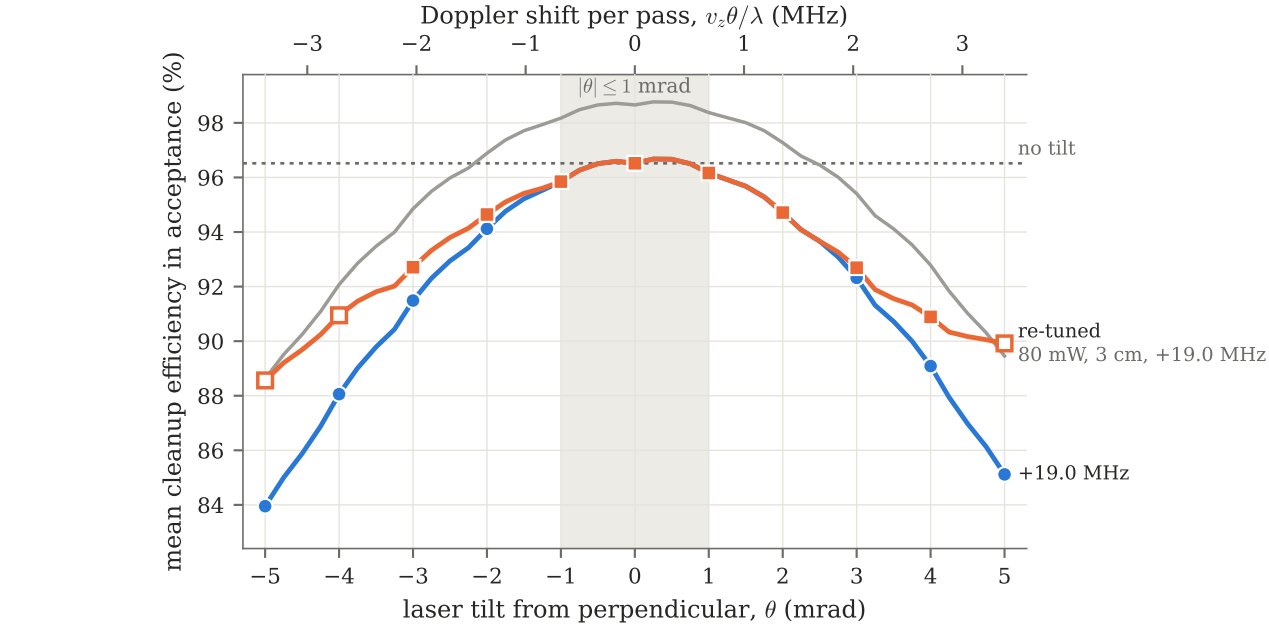


CU1, 50 mW, 2 cm beam ($\sigma_y = 5$ mm), no EOM, start in the 8 SPB-related $J = 1$ levels; mean over $v_x$ (Gaussian, $\sigma = 0.65$ m/s, $|v_x| \le 1$ m/s) and $y_0$ ($\sigma = 10.95$ mm, $|y_0| \le 8.5$ mm). A tilt adds $kv_z \sin\theta$ to every pass, 0.68 MHz/mrad at $v_z = 184$ m/s. Antiparallel passes (retro-reflecting prisms, as in the model): the shift alternates in sign like the transverse one, i.e. an effective $v_x$ shift of $v_z\theta$ = 0.184 m/s per mrad that re-tuning cannot remove. Forward-velocity spread neglected.